



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3445UA0-1



FRIDAY, 16 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

HIGHER TIER
1 hour 30 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	11	
3.	16	
4.	6	
5.	8	
6.	13	
7.	13	
Total	75	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
A Periodic Table is printed on page 20.



Answer **all** questions.

1. Sioned reacted sulfuric acid with zinc oxide (ZnO).
She followed the method below to form zinc sulfate.

1. Pour 50 cm³ of 1.0 mol/dm³ sulfuric acid into a conical flask.
2. Heat the sulfuric acid to 50°C.
3. Add 5 g of zinc oxide to the warmed acid and stir.
4. Repeat step 3 until the zinc oxide is in excess.

- (a) The main hazard in this experiment is that 1.0 mol/dm³ sulfuric acid is an irritant.

- (i) Describe **one** risk associated with using sulfuric acid.

[1]

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- (ii) State a suitable control measure Sioned should carry out to reduce this risk.

[1]

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- (b) Circle the correct formula in each bracket to complete the symbol equation for this reaction.

[2]



- (c) State **two** additional steps that Sioned needs to carry out to obtain pure zinc sulfate crystals.

[2]

1.

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2.

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- (d) Robert has calculated the relative formula mass of zinc oxide to be 97.
Use the Periodic Table on page 20 to explain if he is correct.

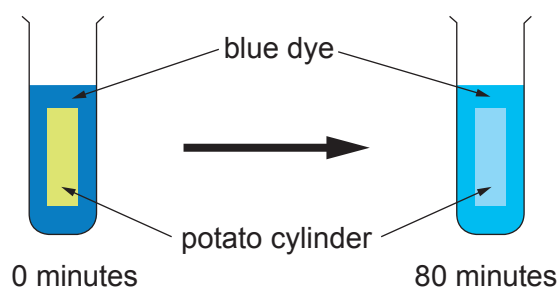
[2]

8

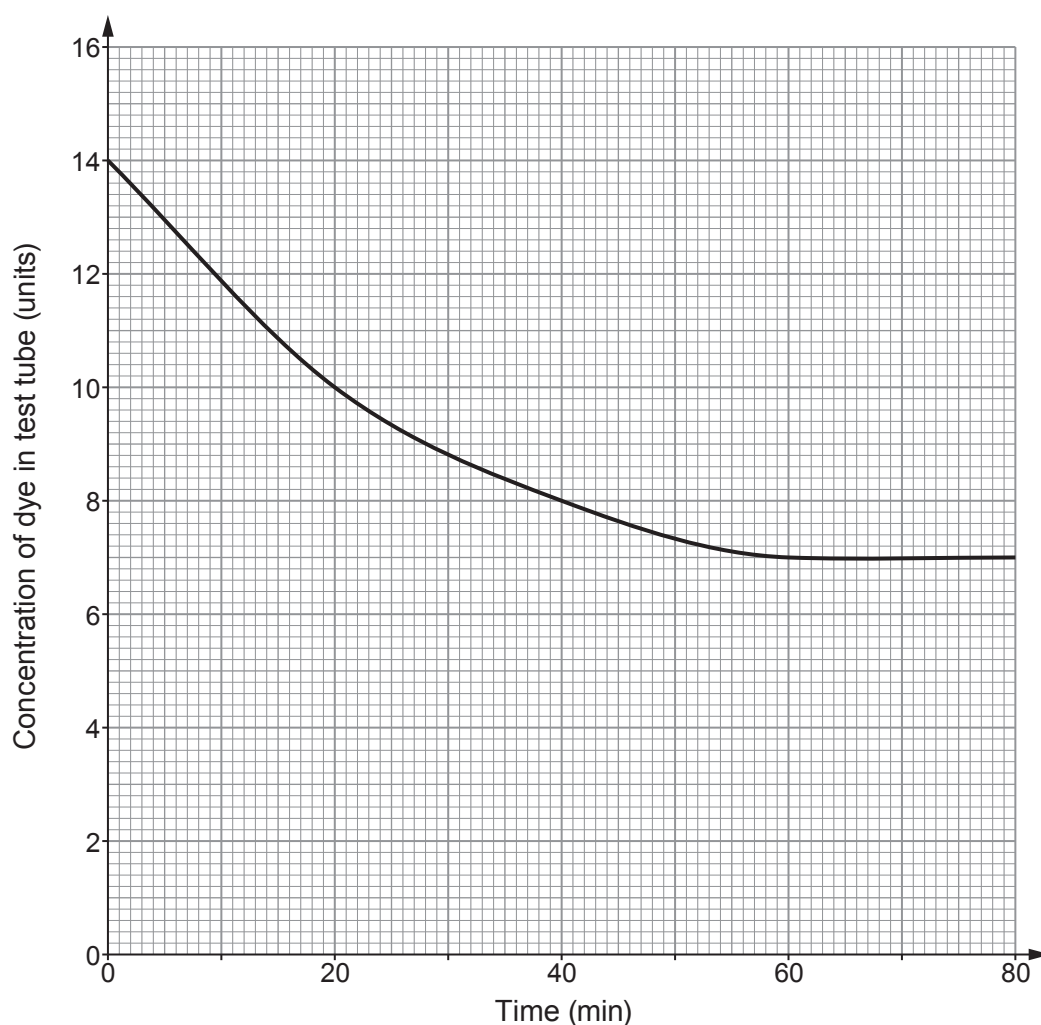
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2. Several groups of students investigated the rate of diffusion of blue dye into potato by the method below.

1. Cut a cylinder of potato.
2. Place the cylinder in a test tube containing a solution of blue dye.
3. Put the test tube in a water bath at 20°C.
4. Measure the concentration of the dye after 20 minutes.
5. Continue to measure the concentration every 20 minutes up to 80 minutes.



The concentration of the dye left in the test tube after each 20-minute interval is shown on the graph below.



- (a) (i) State the dependent variable in this experiment. [1]

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- (ii) State **two** variables that should be controlled. [2]

1.

2.

- (b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

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- (ii) Explain why this change occurred. [2]

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- (c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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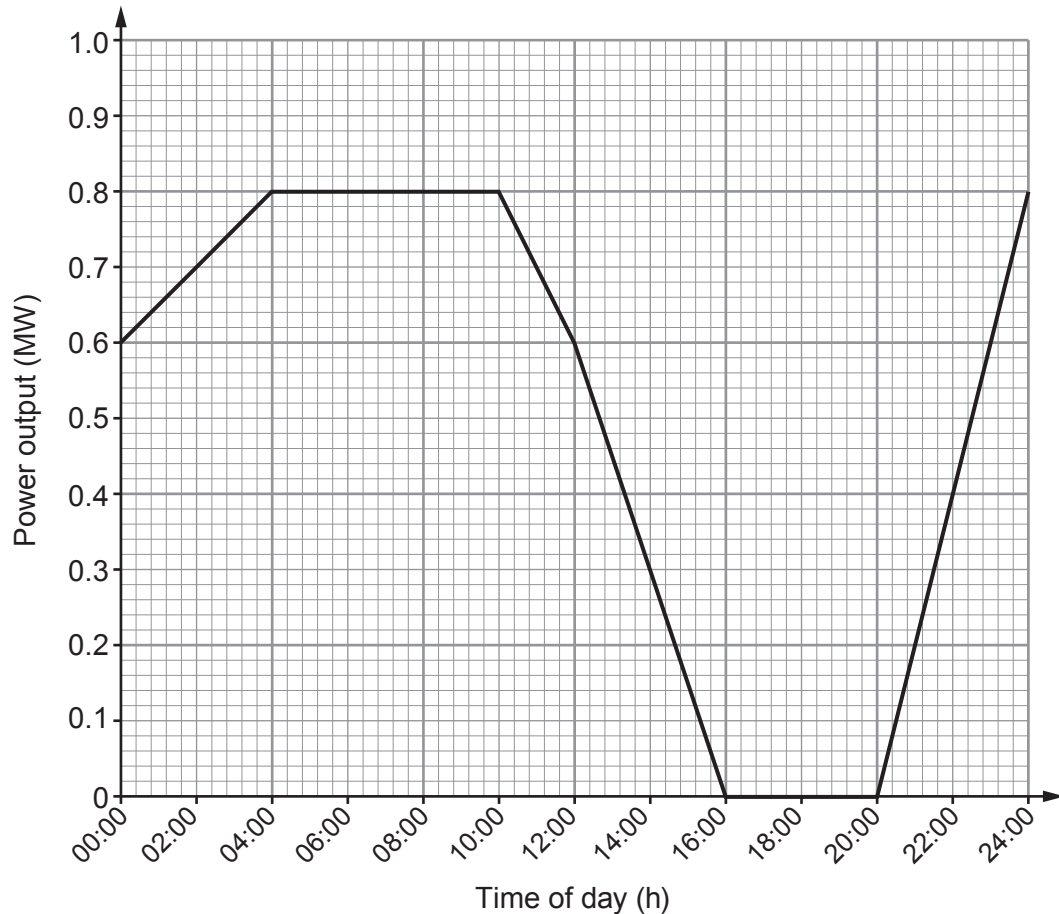
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- (d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]



3. One renewable energy source that large energy suppliers use to generate electricity is wind. The percentage of electricity generated in this way has been gradually increasing over the last few years.

The graph below shows the power output of a wind turbine in one day. This wind turbine has a mean output of 0.6 MW.



- (a) Use the graph and the equation:

$$\text{units generated (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the number of units generated between 04:00 and 10:00.

[4]

(1 MW = 1000 kW)

number of units = kWh



- (b) The table below gives information about generating electricity by wind and a nuclear power station.

	wind turbine	nuclear power station
expected lifetime (years)	20	60
mean power output (MW)	0.60	3 600
land area needed (km ²)	0.55	4.5
cost to commission (£ million)	3	4 000
waste produced	none	radioactive waste
lifetime carbon footprint (gCO ₂ /kWh)	5	5
overall cost of generating electricity (p/kWh)	5.6	2.8

Use the table to answer the following questions.

- (i) Calculate the number of wind turbines that need to be built to provide the same power as the nuclear power station will provide in its lifetime. [2]

number of wind turbines =

- (ii) Calculate the land area needed by a wind farm to produce the same power as one nuclear power station. [3]

land area = km²



- (c) (i) Evaluate the cost effectiveness of the two methods of generating electricity. [3]

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- (ii) Evaluate the environmental impact of the two methods of generating electricity. [2]

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- (d) The lifetime carbon footprints of a wind farm and a nuclear power station are $5\text{gCO}_2/\text{kWh}$.

Suggest how the carbon footprint of a coal-fired power station might compare to this value.

Give a reason for your answer.

[2]

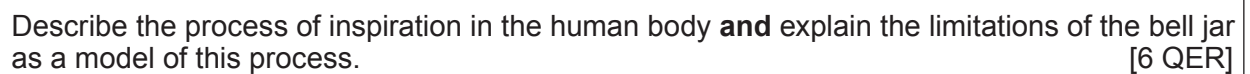
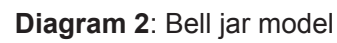
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- Diagram 1: Side view of the chest cavity**



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6



5. Developments in plastic technology have allowed the production of bioplastics using plants as the source of raw material.

The table below compares the production of the same mass of a bioplastic with polystyrene. Polystyrene is made from crude oil.

	Bioplastic	Polystyrene
Energy consumption (kWh)	10.7	22.6
Water used in production (dm ³)	74.6	185
Waste for landfill produced (g)	1.74	2.10
Biodegradable	Yes	No
Breakdown products	carbon dioxide, water and methane	styrene
kgCO ₂ eq/kg produced	0.50	2.2

Use your knowledge and the information in the table to answer the following questions.

- (a) Explain **two** advantages to the environment of using bioplastics. [4]

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- (b) Explain **one** environmental disadvantage of using bioplastics. [2]

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- (c) Manufacturers of this bioplastic claim that it is a carbon-neutral product. State the additional information that is needed to support this claim. [2]

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6. Jamie is investigating series and parallel electrical circuits.

The following equations apply to electrical circuits:

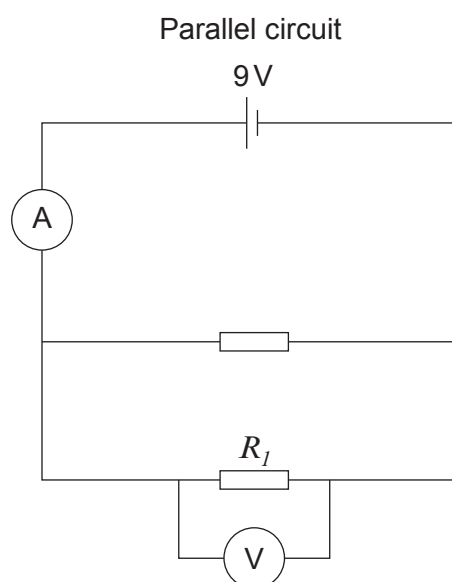
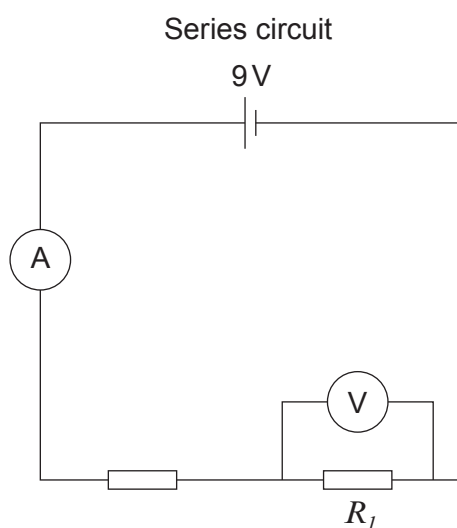
$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

$$\text{power} = \text{voltage} \times \text{current}$$

$$\text{power} = \text{current}^2 \times \text{resistance}$$

$$R_T = R_1 + R_2 \dots\dots\dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \dots\dots\dots$$



- (a) Jamie built the circuits above with identical resistors. In the series circuit the total resistance is 4.5 ohms and the ammeter reading is 2 Amps.

- (i) I. Compare the total resistance in the parallel circuit with the series circuit. [2]

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II. Compare the ammeter reading in the parallel circuit with the series circuit. [1]

(ii) Explain how the voltmeter readings in the two circuits compare. [2]

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(iii) Explain how the power dissipated in resistor R_I in each circuit compares. [2]

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(b) Jamie used identical 2.25Ω resistors in both circuits.

(i) Calculate the power dissipated by R_I in the parallel circuit. [4]

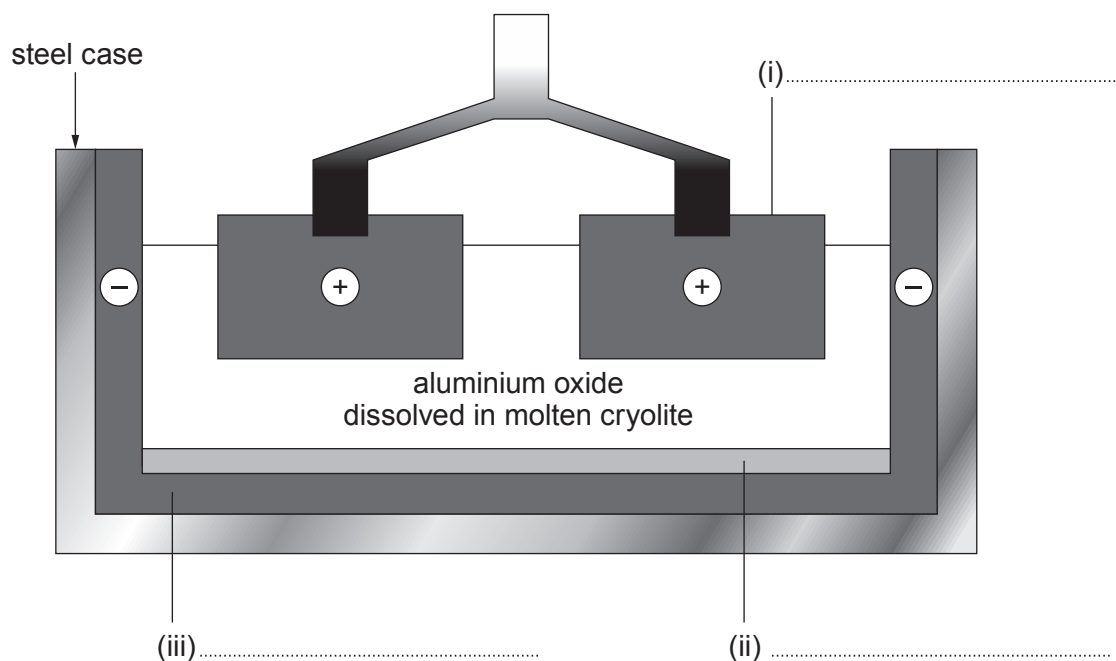
power dissipated = W

(ii) Use your answer above to calculate the power dissipated in resistor R_I in the series circuit. [2]

power dissipated = W



7. The diagram below shows the extraction of aluminium from aluminium oxide by electrolysis.



- (a) Complete the labelling of the diagram above. [3]
- (b) Complete the balanced **symbol** equation for this process. [3]



- (c) Explain, in terms of electrons, what happens at each of the electrodes. [3]

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(d) Over the last 10 years the mass of aluminium recycled in Wales has increased.

(i) Suggest **two** economic benefits of recycling aluminium. [2]

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(ii) Explain **one** environmental benefit of recycling aluminium. [2]

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Symbol	Name	Z
A_r		

atomic number